τριγωνο-ν triangle + μετρία measurement

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what is trig?

Two most basic figures studied in Geometry are the triangle & the circle!

e.g → in Geo you learn that if you know the lengths of 3 sides of a triangle, then the measures of its angles are completely determined & everything else about the triangle is determined.

But for a few "special" triangles Geo. does not tell you how to compute the measures of the angles → given the measures of the sides.

e.g → The measures of the sides of a triangle are

→ 6, 6, & 6 centimeters. $\phi 8.17.25$

Sol. The triangle has 3 equal sides, so its 3 angles are also equal. Since the sum of the angles is $18\phi°$ → the degree-measure of each angle is $\frac{18\phi}{3} = 6\phi°$

Geom. allows you to know this without actually measuring the angles, or even drawing the angles

e.g 2 → The measures of the sides of a triangle are 5, 6, 7 centimeters. What are the measures of its angles?

Sol. You cannot find these angle measures using Geom. The best thing →

you can do is draw the triangle, & measure the angles with a protractor

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→ But how will you know (accurately) what you measured?

e.g 3 → 2 sides of a triangle have length 3 & 4 centimeters & the angle between them is 90° What are the measures of the 3rd side & other two angles.

Geom. tells you → SAS that if you know two sides & an included angle of a triangle → then you ought to be able to find the rest of its measure-ments, in this case → use the Pyth Theorem →

which will tell you the 3^{rd} side of the triangle (has measure 5)
But geom. will not tell you the measures of the angles

e.g → In a certain circle, a central angle of $20°$ cuts off an arc that is 5" long. In the same circle, how long is the arc cut off by a central angle of $40°$?

Sol → You can divide the $40°$ angle into two angles of $20°$. Each of these angles cuts off an arc of length 5", so the arc cut off by the $40°$ angle is $5 + 5 = 10$" long

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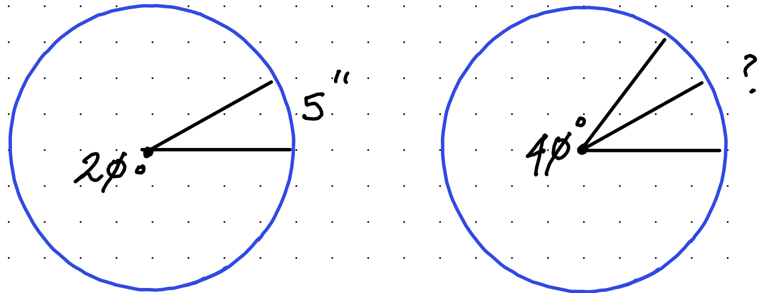

i.e. the double of the central angle, you also double the length of the arc it intercepts

e.g → 5 → In a certain circle, a central angle of 20° determines a chord that is 7 inches long. In the same circle, how long is the chord determined by a central angle of 40°

Sol → you can divide the 40 angle into two angles of 20° angles

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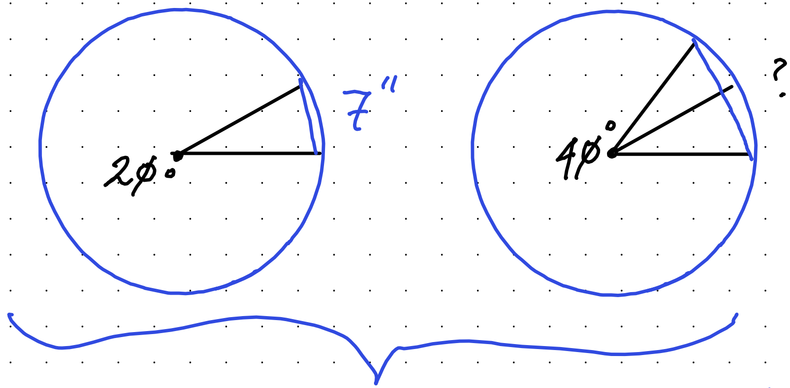

it is not easy to relate the length of the chord determined by the 40° angle to the lengths of the chords of the 20° angles.

Having doubled the angle, you certainly have not doubled the chord.

1. In a circle, suppose you draw any central angle at all → then draw a (2nd) central angle always be longer than the arc of the first?

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Will the chord of the 2nd central angle also be larger than the chord of the first?

2) What theorem in geom. guarantees that the chord of a 40° angle is less than double the chord of a 20° angle

3). Suppose you draw any central angle, then double it. Will the chord of the double angle always be less than twice the chord of the original central angle?

Trig. & Geom. tell that any two equal arcs in the same circle have equal chords; i.e → if you know the measurement of the Arc then the length of the chord is determined. But, except in special circumstances, geom. does not give enough tools to calculate the length of the chord knowing the measure of the arc.

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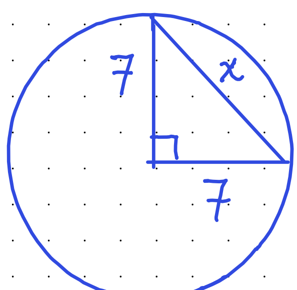
e.g → In a circle of radius 7, how long is the chord of an arc 90°?

Sol. If you draw radii to the endpoints of the chord → you will need a isoseles right triangle:



use the Pythagorean Theorem to find the length of the chord →

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If this length is x, then

$$7^2 + 7^2 = x^2$$

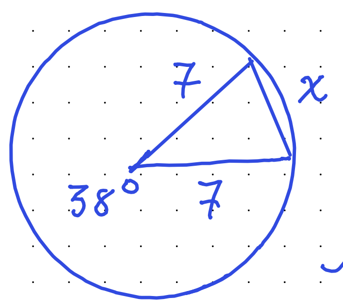
so →

$$49 + 49 = x^2$$

$$\sqrt{98} = x$$

e.g 7 → In a circle of Radius, 7 how long is the chord of an arc of 38°?

Sol → Geom. does not give us the tools to solve this → draw a triangle



But you cannot find the 3rd side of this triangle using only geom. → it does illustrate the close connection between measurements in a triangle & measurements in a circle.

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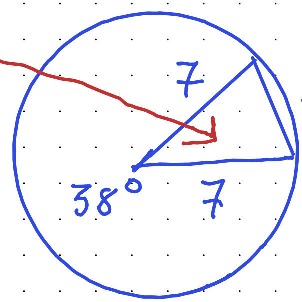
ex 1. What theorem from Geom. guarantees that the triangle ~ is completely determined?

2. The triangle is isosceles.



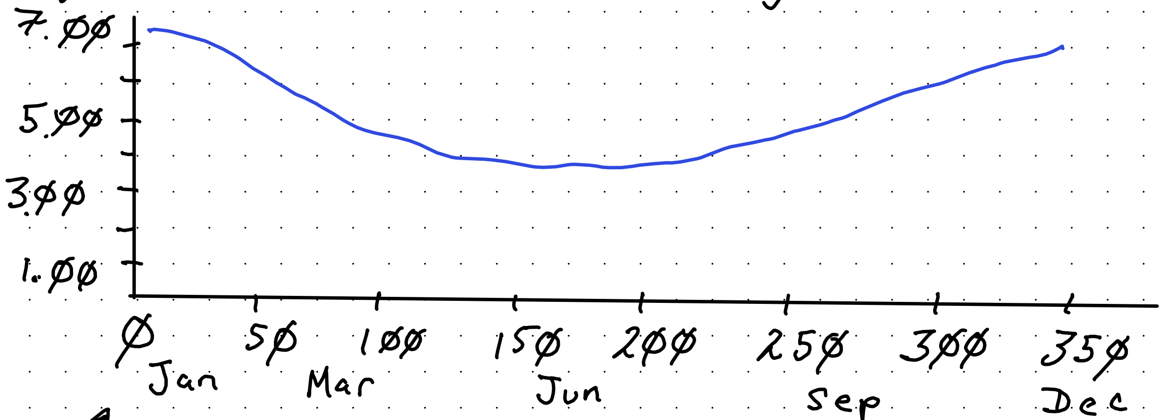
calculate the measure of the two missing angles?

Trig will help you solve all these kinds of problems. However, trig is more than just an extension of geom.

Applications of trig abound in many →

→ many branches of science!   08.17.2025

e.g → & look at any pendulum as it swings,
If you look closely, → you will see
that the weight travels very slowly @ either
end of its path, & picks up speed as it gets
toward the middle. It travels fastest
during the middle of its journey



↑ some graph → shows (corrected for
daylight

@ a certain latitude for Wednesdays Ø8, 17
in the year 1995 2Ø25
The data points have been joined by
a smooth curve to make a continuous
graph. (over the entire year)
you would expect this curve to be
essentially the same year after year
However, neither geom. nor algebra can give
you a formula for this curve.
.. you will see how trig allows you to
describe it mathematically. Trig allows you
to investigate any periodic phenomenon
-- any physical motion or change
that repeats itself.

Right Triangles

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let's use right triangles for now →

get a good understanding of Right Triangles.

when describing triangles → you already know that one angle measures $90°$

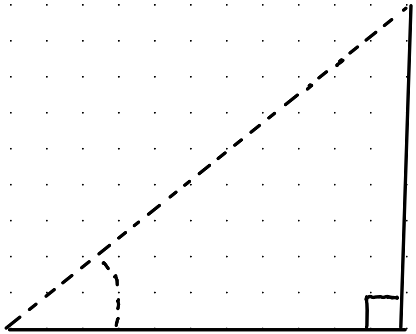
a) the lengths of the two legs;

b) the lengths of one leg & the Hype.;

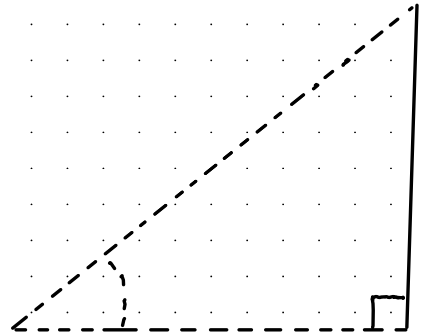
c) the length of one leg & the measure of one acute angle;

d) the length of the Hype & the measure of one acute angle

↓



a



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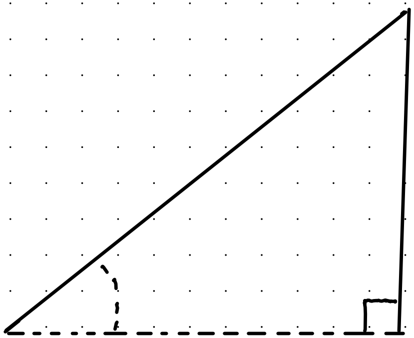
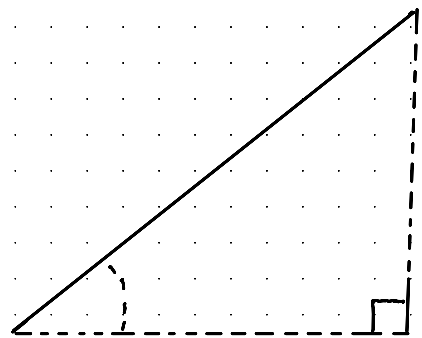


B



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suppose that you want to know the lengths of all sides of the triangle

For (a) & (b) you need algebra & Geom. For (c) & (d) algebriac expr. do not usually suffice

The Pythagorean Theorem 08.17.2025

look at the first chief geometric tool which allows you to solve cases (a) & (b)

This is the famous Pythagorean Theorem
you seperate the Pyth. theorem into two statements →

① If a & b are the lengths of the legs of a right triangle, & c is the length of its hypotenuse, then →

$$a^2 + b^2 = c^2$$

(II) If the positive numbers → a, b & c satisfy →

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$a^2 + b^2 = c^2$, then a triangle with these side lengths has a right angle opp. the side with length c^2

These two statements are converses of each other. They look similar, but a careful reading will show that they say completely different things about triangles.

In the first statement → you say you Know something about an angle, of a Triangle ($90°$) Right Triangle & conclude $\Rightarrow$ that a certain relationship holds among the sides.

In the 2nd statement → 08.17.2025
you know something about the sides of the triangle & conclude something about the angles (that one of them is a right angle)

The Pyth theorem will allows to reconstruct a triangle, given two legs or a leg & the hype.
This is because you can find, using this info, the lengths of all three sides of the triangle. As you know from geom. this completely determines the triangle.

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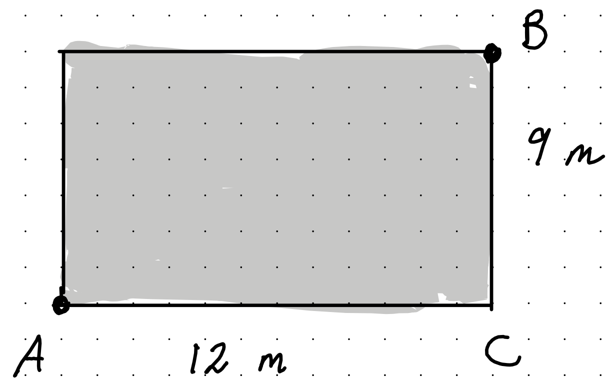
eg. → in Oxford → there are some-times lawns occupying rectanglar lots near the intersection of two roads



going back to statement 2

If the positive numbers a, b, & c satisfy → $a^2 + b^2 = c^2$, then there exists a triangle with sides a, b, & c & this triangle has a right angle opposite the side with length c

This statement includes → $\phi 8.17.2\phi 25$

e.g. the fact that if $a^2 + b^2 = c^2$,

then $a + b > c$

How much further must the students walk than the professors in going from $A \rightarrow B$

Sol → Triangle ABC is a right triangle, so statement I of the Pythagorean theorem applies:

$$AB^2 = AC^2 + BC^2$$

$$= 12^2 + 9^2$$

$$= 144 + 81 = 255$$

So → $AB = 15$ → how far the prof. walks

the students must walk the distance

$$AC + CB = 12 + 9 = 21$$

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